Muddana Ahalya

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Github Linkedin

Objective

Recent B.Tech graduate in Information Technology with a strong foundation in programming, databases, software development, and machine learning. Seeking an entry-level opportunity to apply my technical skills, contribute to real-world projects, and grow as a technology professional while delivering practical and scalable solutions.

Work Experience

AI-ML Virtual Intern Dec 2024 - Feb 2025

- Gained hands-on experience with supervised and unsupervised learning, deep learning, and neural networks.
- Implemented end-to-end Machine learning workflow including data preparation, feature engineering, model training, and evaluation.
- Explored AWS services such as Amazon SageMaker, Bedrock, and Forecast to build, deploy, and scale Machine learning models.

Projects

Smart Soil Nutrient and Moisture Monitoring System *Jan 2026 - may 2026*

IoT, Arduino, Modbus RS485, Node.js, Express.js, Python, Scikit-Learn, Chart.js

- Developed an IoT-driven monitoring system integrating Modbus RS485 NPK sensors and capacitive moisture sensors with Arduino to capture real-time soil nutrient telemetry.
- Integrated a Node.js and Express backend with a Python Scikit-Learn Decision Tree model to process live serial data and generate real-time, hourly, and daily crop suitability recommendations.
- Designed a responsive web dashboard using Chart.js to visualize real-time soil health metrics and enable automated CSV dataset logging and export for precision farming.

VecTorium: Hybrid Deep Learning Ensemble for Brain Tumor Classification *Oct 2025 – Nov 2025*

Python, TensorFlow, Keras, EfficientNetB3, ResNet50, VGG16, OpenCV, Scikit-Learn, NumPy, Matplotlib

- Engineered a custom deep learning ensemble model (**VecTorium**) integrating frozen pre-trained CNN backbones (EfficientNetB3, ResNet50, VGG16) with GlobalAveragePooling2D and feature concatenation for 4-class MRI brain tumor classification.
- Implemented real-time image preprocessing, data augmentation (*ImageDataGenerator*), dynamic learning rate decay (*ReduceLROnPlateau*), and *EarlyStopping* callbacks to optimize training and prevent overfitting.
- Evaluated model performance using classification metrics, training/validation loss curves, and confusion matrices, achieving **96.73% validation accuracy** and **99.49% training accuracy**.

Education

Bachelor's of Technology 2022 – 2026
QIS College of Engineering and Technology, Ongole, Andhra Pradesh Information Technology (GPA: 8.3)

Intermediate Education 2020 – 2022
Sri Chaitanya Junior College, Addanki, Andhra Pradesh MPC (85.7%)

Secondary Education 2020
ZPHS High School, Kondamanjuluru, Andhra Pradesh SSC (100%)

Technical Skills

Programming	Python(Intermediate), SQL(Intermediate), Java(Basic), HTML(Intermediate), CSS (Intermediate), Javascript(Intermediate)
Libraries	Pandas (Intermediate), NumPy (Intermediate), Scikit-learn (Intermediate), PyTorch (Basic)
Tools	Git/GitHub(Intermediate), Jupyter Notebook(Intermediate), VS Code(Intermediate), Arduino UNO(Intermediate), Excel(Intermediate)

Certificates

- **Spring and Angular Full Stack Developer(Infosys)** June 11, 2025
- **Python Full stack developer(Eduskills)** October-December 2024
- **AWS Academy Cloud Foundations (Amazon)** April 2025
- **python programming(codetantra)** April 2025

Coursework & Activities

Relevant Coursework	Data Structures (Basics), Data base management system(DBMS), Object oriented programming(OOPS), Machine Learning
Activities	Participated in hackathons and technical activities, strengthening teamwork, problem-solving, communication, time management, and collaboration skills while working under tight deadlines.